

Diagnostic Accuracy of IDX-DR for Detecting Diabetic Retinopathy: A Systematic Review and Meta-Analysis



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• **PURPOSE:** Diabetic retinopathy (DR) is a leading cause of vision loss worldwide, making early detection critical to prevent blindness. IDX-DR, an FDA-approved autonomous artificial intelligence (AI) system, has emerged as an innovative solution to improve access to DR screen-

ing. This systematic review and meta-analysis aimed to evaluate the diagnostic accuracy of IDX-DR in detecting diabetic retinopathy.

• **DESIGN:** Systematic review and meta-analysis.

• **METHODS:** A comprehensive literature search was conducted across PubMed, Embase, Scopus and Web of Science, identifying studies published through October 5, 2024. Studies involving adult patients with Type 1 or Type 2 diabetes and reporting diagnostic metrics such as sensitivity and specificity were included. The primary outcomes were pooled sensitivity and specificity of IDX-DR. A bivariate random-effects model was used for meta-analysis, and summary receiver operating characteristic (SROC) curves were generated to assess diagnostic performance. Statistical analyses were performed using MetaDisc software version 2.0.

• **RESULTS:** Thirteen studies involving 13,233 participants met the inclusion criteria. IDX-DR's pooled sensitivity was 0.95 (95% CI: 0.82-0.99), and its pooled specificity was 0.91 (95% CI: 0.84-0.95). The SROC curve confirmed IDX-DR's high diagnostic accuracy in detecting diabetic retinopathy across various clinical environments. The AUC value of 0.95 demonstrated high sensitivity and specificity, indicating a robust diagnostic performance for IDX-DR in detecting diabetic retinopathy.

• **CONCLUSION:** IDX-DR is a highly effective diagnostic tool for diabetic retinopathy screening, with robust sensitivity and good specificity. Its integration into clinical practice, especially in resource-limited settings, can potentially improve early detection and reduce vision loss. However, careful implementation is needed to address challenges such as over-diagnosis and ensure the tool complements clinical judgment. Future studies should explore the long-term impacts of AI-based screening and address ethical considerations surrounding its use. (Am J Ophthalmol 2025;273: 192–204. © 2025 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.)

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